RAG Chat Test Suite - 60 Questions

Database: ralli_agrotech_2_db (dump 03-Nov-2026). 30 simple + 30 critical questions. Ask each question in the chat, run the validation SQL in MySQL 8 and compare. Expected results are the snapshot of the dump with today = 2026-11-04; C01, C05 and C22 depend on CURDATE() so they change as days pass. Equal-ranked rows may appear in a different order.

Simple questions (30)

S01 · Simple	How many karigars are there in total?
VALIDATION SQL (MySQL 8)	`SELECT COUNT(*) AS total_karigars FROM karigars`
EXPECTED RESULT	`total_karigars=35`
CHECKS	-

S02 · Simple	How many customer (job) orders are there?
VALIDATION SQL (MySQL 8)	`SELECT COUNT(*) AS total_orders FROM customerorders`
EXPECTED RESULT	`total_orders=579`
CHECKS	-

S03 · Simple	How many order line items are there in total?
VALIDATION SQL (MySQL 8)	`SELECT COUNT(*) AS total_order_items FROM customerorderitems`
EXPECTED RESULT	`total_order_items=2950`
CHECKS	'line items' must not trigger chart mode

S04 · Simple	What is the total ordered quantity across all order items?
VALIDATION SQL (MySQL 8)	`SELECT SUM(ord_qty_actual) AS total_ordered_qty FROM customerorderitems`
EXPECTED RESULT	`total_ordered_qty=9388`
CHECKS	-

S05 · Simple	How many orders are still in Processing status?
VALIDATION SQL (MySQL 8)	`SELECT COUNT(*) AS processing_orders FROM customerorders WHERE issue_to_karigar='Processing'`
EXPECTED RESULT	`processing_orders=37`
CHECKS	-

S06 · Simple	How many order items are Complete and how many are Processing?
VALIDATION SQL (MySQL 8)	`SELECT issue_to_karigar_status AS status, COUNT(*) AS items, SUM(ord_qty_actual) AS qty FROM customerorderitems GROUP BY issue_to_karigar_status ORDER BY items DESC`
EXPECTED RESULT	`status=Complete | items=2779 | qty=8919` `status=Processing | items=171 | qty=469`
CHECKS	-

S07 · Simple	List all karigar names with their codes.
VALIDATION SQL (MySQL 8)	`SELECT kid, kname FROM karigars ORDER BY kname`
EXPECTED RESULT	kid=AKD | kname=ABHISHEK DAS kid=AR | kname=AMAL ROY kid=AD | kname=ANIL KUMAR DHAR kid=AT | kname=ASISH TARAFDAR kid=XX | kname=AUTO kid=BP | kname=BALARAM PORE kid=BLJ | kname=BHAGYA LAXMI JEWELLERS kid=BM | kname=BIBHAS MAJI kid=BB | kname=BISWANATH BERA kid=CR | kname=CHAND RATAN SONI ... (35 rows total)
CHECKS	-

S08 · Simple	What is the name of the karigar with code BLJ?
VALIDATION SQL (MySQL 8)	`SELECT kid, kname FROM karigars WHERE kid='BLJ'`
EXPECTED RESULT	kid=BLJ | kname=BHAGYA LAXMI JEWELLERS
CHECKS	code lookup

S09 · Simple	What is the mobile number of Kamal Paul?
VALIDATION SQL (MySQL 8)	`SELECT kname, mobile FROM karigars WHERE kname LIKE '%KAMAL PAUL%'`
EXPECTED RESULT	kname=KAMAL PAUL | mobile=6290306850
CHECKS	name LIKE lookup

S10 · Simple	What is the GSTIN of Bhagya Laxmi Jewellers?
VALIDATION SQL (MySQL 8)	`SELECT kname, gstin FROM karigars WHERE kname LIKE '%BHAGYA LAXMI%'`
EXPECTED RESULT	kname=BHAGYA LAXMI JEWELLERS | gstin=19AHVPS9192D1ZE
CHECKS	name LIKE lookup

S11 · Simple	How many items are in job order 11705351 and what is their total ordered quantity?
VALIDATION SQL (MySQL 8)	`SELECT o.jo_no, COUNT(*) AS items, SUM(c.ord_qty_actual) AS total_qty FROM customerorderitems c JOIN customerorders o ON o.id=c.order_id WHERE o.jo_no='11705351' GROUP BY o.jo_no`
EXPECTED RESULT	jo_no=11705351 | items=3 | total_qty=17
CHECKS	job_no vs order_id join

S12 · Simple	Show the item codes, descriptions and delivery dates for job order 11705351.
VALIDATION SQL (MySQL 8)	`SELECT TRIM(c.item_code) AS item_code, c.description, c.delivery_date FROM customerorderitems c JOIN customerorders o ON o.id=c.order_id WHERE o.jo_no='11705351' ORDER BY c.sl_no+0`
EXPECTED RESULT	item_code=50O6FDGATGAA00 | description=NECKLACE | delivery_date=2026-08-11 item_code=50O6FD0ATGBA00 | description=PENDENT WITH CHAIN | delivery_date=2026-08-11 item_code=50O6FDDATABA00 | description=DROP EARRING | delivery_date=2026-08-11
CHECKS	job lookup + TRIM item_code

S13 · Simple	How many distinct jobs have been issued to karigars?
VALIDATION SQL (MySQL 8)	SELECT COUNT(DISTINCT job_no) AS issued_jobs FROM issuetokarigaritems
EXPECTED RESULT	issued_jobs=566
CHECKS	-

S14 · Simple	What is the total quantity issued to karigars and how much is still pending (balance)?
VALIDATION SQL (MySQL 8)	SELECT SUM(qty) AS issued_qty, SUM(bal_qty) AS pending_qty FROM issuetokarigaritems
EXPECTED RESULT	issued_qty=9165 | pending_qty=583
CHECKS	-

S15 · Simple	How many quality check vouchers have been created?
VALIDATION SQL (MySQL 8)	SELECT COUNT(DISTINCT qc_voucher) AS qc_vouchers FROM qualitychecks
EXPECTED RESULT	qc_vouchers=581
CHECKS	-

S16 · Simple	How many quality check items are still in Processing status?
VALIDATION SQL (MySQL 8)	SELECT COUNT(*) AS qc_items_processing FROM qualitycheckitems WHERE status='Processing'
EXPECTED RESULT	qc_items_processing=622
CHECKS	-

S17 · Simple	How many finished product receive vouchers are there?
VALIDATION SQL (MySQL 8)	SELECT COUNT(*) AS fpr_vouchers FROM finishproductreceivedentries
EXPECTED RESULT	fpr_vouchers=894
CHECKS	-

S18 · Simple	What is the total net weight received in finished product vouchers?
VALIDATION SQL (MySQL 8)	SELECT ROUND(SUM(voucher_net_wt),2) AS total_net_wt FROM finishproductreceivedentries
EXPECTED RESULT	total_net_wt=152600.27
CHECKS	-

S19 · Simple	How many finished goods in/out vouchers exist and what is their date range?
VALIDATION SQL (MySQL 8)	SELECT COUNT(*) AS vouchers, MIN(voucher_dt) AS first_date, MAX(voucher_dt) AS last_date FROM f_g_in_out__h_e_a_d_e_r
EXPECTED RESULT	vouchers=574 | first_date=2026-08-30 | last_date=2026-10-26
CHECKS	-

S20 · Simple	How many products are there in the product master?
VALIDATION SQL (MySQL 8)	`SELECT COUNT(*) AS products FROM products`
EXPECTED RESULT	`products=20130`
CHECKS	-

S21 · Simple	How many metal issue entries are there and what is the total metal weight issued?
VALIDATION SQL (MySQL 8)	`SELECT COUNT(*) AS entries, ROUND(SUM(weight),3) AS total_weight FROM metalissueentries`
EXPECTED RESULT	`entries=106 | total_weight=87247.649`
CHECKS	-

S22 · Simple	What purities are available in the purity master?
VALIDATION SQL (MySQL 8)	`SELECT purity_id, metal_id, purity FROM metalpurities ORDER BY metal_id, purity`

EXPECTED RESULT	`purity_id=43 | metal_id=1 | purity=37.5`
	`purity_id=42 | metal_id=1 | purity=58.3`
	`purity_id=41 | metal_id=1 | purity=58.5`
	`purity_id=1 | metal_id=1 | purity=75.0`
	`purity_id=2 | metal_id=1 | purity=91.6`
	`purity_id=3 | metal_id=1 | purity=91.7`
	`purity_id=44 | metal_id=1 | purity=91.8`
	`purity_id=4 | metal_id=1 | purity=99.5`
	`purity_id=40 | metal_id=1 | purity=99.52`
	`purity_id=5 | metal_id=1 | purity=99.6`
	`... (36 rows total)`

CHECKS	-

S23 · Simple	How many rework or rejection records are there?
VALIDATION SQL (MySQL 8)	`SELECT COUNT(*) AS rejection_records FROM customers`
EXPECTED RESULT	`rejection_records=310`
CHECKS	-

S24 · Simple	What are the rejection reasons and how many records does each have?
VALIDATION SQL (MySQL 8)	`SELECT reason, COUNT(*) AS records FROM customers GROUP BY reason ORDER BY records DESC`

EXPECTED RESULT	`reason=DT REWORK | records=305`
	`reason=OTHER | records=4`
	`reason=PURITY REJECTION | records=1`

CHECKS	-

S25 · Simple	Which karigar has received the most finished product vouchers?
VALIDATION SQL (MySQL 8)	`SELECT k.kname, COUNT(*) AS vouchers FROM finishproductreceivedentries e JOIN karigars k ON k.id=e.karigar_id GROUP BY k.id, k.kname ORDER BY vouchers DESC LIMIT 1`
EXPECTED RESULT	`kname=BHAGYA LAXMI JEWELLERS | vouchers=114`
CHECKS	-

S26 · Simple	What is the total stone weight in finished product items?
VALIDATION SQL (MySQL 8)	`SELECT ROUND(SUM(stone_wt),3) AS total_stone_wt FROM finishproductreceivedentryitems`
EXPECTED RESULT	`total_stone_wt=72.41`
CHECKS	-

S27 · Simple	How many 18KT and 22KT order items are there?
VALIDATION SQL (MySQL 8)	`SELECT kt, COUNT(*) AS items FROM customerorderitems GROUP BY kt ORDER BY kt`
EXPECTED RESULT	`kt=18 | items=561` `kt=22 | items=2389`
CHECKS	-

S28 · Simple	What is the description and ordered quantity for item code 50O6FDGATGAA00?
VALIDATION SQL (MySQL 8)	`SELECT TRIM(item_code) AS item_code, description, SUM(ord_qty_actual) AS ordered_qty FROM customerorderitems WHERE TRIM(item_code)='50O6FDGATGAA00' GROUP BY TRIM(item_code), description`
EXPECTED RESULT	`item_code=50O6FDGATGAA00 | description=NECKLACE | ordered_qty=1`
CHECKS	TRIM item_code

S29 · Simple	What are the standard weight and labour charge of product 5100222KAAAA00?
VALIDATION SQL (MySQL 8)	`SELECT TRIM(item_code) AS item_code, description, standard_wt, lab_charge FROM products WHERE TRIM(item_code)='5100222KAAAA00'`
EXPECTED RESULT	`item_code=5100222KAAAA00 | description=22KT - HALF SET | standard_wt=80.89 | lab_charge=41`
CHECKS	products TRIM join

S30 · Simple	How much metal in total has been received from TITAN COMPANY LIMITED?
VALIDATION SQL (MySQL 8)	`SELECT cust_name, ROUND(SUM(weight),3) AS total_received_weight FROM metalreceiveentries WHERE cust_name LIKE '%TITAN%' GROUP BY cust_name`
EXPECTED RESULT	`cust_name=TITAN COMPANY LIMITED | total_received_weight=142789.403`
CHECKS	-

Critical questions (30)

C01 · Critical	Which karigars hold the most overdue open order items (Processing and delivery date before today)?

VALIDATION SQL (MySQL 8)	`SELECT SUBSTRING_INDEX(TRIM(c.kid),' - ',1) AS kid, k.kname, COUNT(*) AS overdue_items, SUM(c.ord_qty_actual) AS overdue_qty`
	`FROM customerorderitems c LEFT JOIN karigars k ON k.kid=SUBSTRING_INDEX(TRIM(c.kid),' - ',1)`
	`WHERE c.issue_to_karigar_status='Processing' AND c.delivery_date < CURDATE()`
	`GROUP BY SUBSTRING_INDEX(TRIM(c.kid),' - ',1), k.kname ORDER BY overdue_qty DESC, overdue_items DESC LIMIT 10`

EXPECTED RESULT	`kid=KP | kname=KAMAL PAUL | overdue_items=3 | overdue_qty=109`
	`kid=SM | kname=SANTU KUMAR MANNA | overdue_items=3 | overdue_qty=61`
	`kid=AKD | kname=ABHISHEK DAS | overdue_items=5 | overdue_qty=16`
	`kid=BLJ | kname=BHAGYA LAXMI JEWELLERS | overdue_items=3 | overdue_qty=11`
	`kid=MDS | kname=HMR JEWELS | overdue_items=2 | overdue_qty=10`
	`kid=SK | kname=SUBHAMAY KARMAKAR | overdue_items=1 | overdue_qty=10`
	`kid=SGJ | kname=SHRI GANNAYAK JEWELLERS | overdue_items=1 | overdue_qty=9`
	`kid=SMS | kname=SAMIR SAMANTA | overdue_items=1 | overdue_qty=2`
	`kid=SD | kname=SAMIR DHARA AND CO. | overdue_items=1 | overdue_qty=2`
	`kid=SKS | kname=SAROJ KUMAR SAMANTA | overdue_items=1 | overdue_qty=1`

CHECKS	overdue + dirty kid normalisation

C02 · Critical	What is the total quantity, weight and value of order items that are still open (Processing)?
VALIDATION SQL (MySQL 8)	`SELECT COUNT(*) AS open_items, SUM(ord_qty_actual) AS open_qty, ROUND(SUM(total_wt),2) AS open_weight, ROUND(SUM(total_value),2) AS open_value FROM customerorderitems WHERE issue_to_karigar_status='Processing'`
EXPECTED RESULT	`open_items=171 | open_qty=469 | open_weight=9448.12 | open_value=222581.54`
CHECKS	open definition (Processing)

C03 · Critical	What are the top 5 karigars by pending (unreturned) issued quantity?
VALIDATION SQL (MySQL 8)	`SELECT k.kname, i.kid, SUM(i.bal_qty) AS pending_qty, SUM(i.qty) AS issued_qty FROM issuetokarigaritems i JOIN karigars k ON k.kid=i.kid GROUP BY i.kid, k.kname ORDER BY pending_qty DESC LIMIT 5`

EXPECTED RESULT	`kname=ABHISHEK DAS | kid=AKD | pending_qty=98 | issued_qty=1822`
	`kname=BHAGYA LAXMI JEWELLERS | kid=BLJ | pending_qty=95 | issued_qty=2409`
	`kname=SAMIR DHARA AND CO. | kid=SD | pending_qty=80 | issued_qty=587`
	`kname=MISMATCH | kid=MM | pending_qty=57 | issued_qty=90`
	`kname=KAMAL PAUL | kid=KP | pending_qty=36 | issued_qty=318`

CHECKS	bal_qty pending

C04 · Critical	What share of the total pending issued quantity is held by the top 3 karigars?
VALIDATION SQL (MySQL 8)	`SELECT SUM(t.pending) AS top3_pending, (SELECT SUM(bal_qty) FROM issuetokarigaritems) AS total_pending, ROUND(100.0*SUM(t.pending)/(SELECT SUM(bal_qty) FROM issuetokarigaritems),1) AS top3_share_pct FROM (SELECT kid, SUM(bal_qty) AS pending FROM issuetokarigaritems GROUP BY kid ORDER BY pending DESC LIMIT 3) t`
EXPECTED RESULT	`top3_pending=273 | total_pending=583 | top3_share_pct=46.8`
CHECKS	share % of top-3

C05 · Critical	Which karigars have overdue pieces, i.e. balance quantity still pending after the delivery date has passed?
VALIDATION SQL (MySQL 8)	`SELECT k.kname, i.kid, COUNT(*) AS overdue_lines, SUM(i.bal_qty) AS overdue_pending_qty FROM issuetokarigaritems i JOIN karigars k ON k.kid=i.kid WHERE i.bal_qty>0 AND i.delivery_date < CURDATE() GROUP BY i.kid, k.kname ORDER BY overdue_pending_qty DESC LIMIT 10`

EXPECTED RESULT	`kname=BHAGYA LAXMI JEWELLERS | kid=BLJ | overdue_lines=60 | overdue_pending_qty=93`
	`kname=ABHISHEK DAS | kid=AKD | overdue_lines=57 | overdue_pending_qty=77`
	`kname=SAMIR DHARA AND CO. | kid=SD | overdue_lines=39 | overdue_pending_qty=73`
	`kname=MISMATCH | kid=MM | overdue_lines=46 | overdue_pending_qty=57`
	`kname=KAMAL PAUL | kid=KP | overdue_lines=17 | overdue_pending_qty=33`
	`kname=AMAL ROY | kid=AR | overdue_lines=5 | overdue_pending_qty=29`
	`kname=SHRI GANNAYAK JEWELLERS | kid=SGJ | overdue_lines=23 | overdue_pending_qty=28`
	`kname=SUBHAMAY KARMAKAR | kid=SK | overdue_lines=18 | overdue_pending_qty=27`
	`kname=SAMIR SAMANTA | kid=SMS | overdue_lines=11 | overdue_pending_qty=17`
	`kname=JITEN PAUL | kid=JP | overdue_lines=12 | overdue_pending_qty=16`

CHECKS	overdue at karigar

C06 · Critical	Which jobs were issued to karigars but never received as finished product, and how many per karigar?
VALIDATION SQL (MySQL 8)	SELECT k.kname, COUNT(DISTINCT i.job_no) AS jobs_not_received, SUM(i.qty) AS issued_qty FROM issuetokarigaritems i JOIN karigars k ON k.kid=i.kid WHERE NOT EXISTS (SELECT 1 FROM finishproductreceivedentryitems f WHERE f.job_no=i.job_no) GROUP BY i.kid, k.kname ORDER BY jobs_not_received DESC LIMIT 10
EXPECTED RESULT	kname=ABHISHEK DAS | jobs_not_received=32 | issued_qty=32 kname=BALARAM PORE | jobs_not_received=11 | issued_qty=11 kname=SAMIR DHARA AND CO. | jobs_not_received=11 | issued_qty=11 kname=BIBHAS MAJI | jobs_not_received=7 | issued_qty=7 kname=MISMATCH | jobs_not_received=7 | issued_qty=8 kname=KAMAL PAUL | jobs_not_received=6 | issued_qty=9 kname=HMR JEWELS | jobs_not_received=6 | issued_qty=6 kname=SHRI GANNAYAK JEWELLERS | jobs_not_received=6 | issued_qty=6 kname=JITEN PAUL | jobs_not_received=5 | issued_qty=5 kname=SUBHAMAY KARMAKAR | jobs_not_received=5 | issued_qty=5
CHECKS	NOT EXISTS across stages

C07 · Critical	What is the loss weight and loss percentage per karigar on finished product vouchers? Rank the highest loss percentage first.
VALIDATION SQL (MySQL 8)	SELECT k.kname, ROUND(SUM(e.voucher_net_wt),2) AS net_wt, ROUND(SUM(e.voucher_loss),2) AS loss_wt, ROUND(100.0*SUM(e.voucher_loss)/SUM(e.voucher_net_wt),2) AS loss_pct FROM finishproductreceivedentries e JOIN karigars k ON k.id=e.karigar_id GROUP BY k.id, k.kname ORDER BY loss_pct DESC
EXPECTED RESULT	kname=KAMAL PAUL | net_wt=13104.5 | loss_wt=500.47 | loss_pct=3.82 kname=CHAND RATAN SONI | net_wt=484.47 | loss_wt=18.41 | loss_pct=3.8 kname=GITA DHAR | net_wt=1250.69 | loss_wt=46.24 | loss_pct=3.7 kname=SUBHAMAY KARMAKAR | net_wt=7680.68 | loss_wt=242.34 | loss_pct=3.16 kname=ABHISHEK DAS | net_wt=13332.68 | loss_wt=363.78 | loss_pct=2.73 kname=BHAGYA LAXMI JEWELLERS | net_wt=18840.22 | loss_wt=510.99 | loss_pct=2.71 kname=BALARAM PORE | net_wt=8624.97 | loss_wt=216.51 | loss_pct=2.51 kname=HMR JEWELS | net_wt=11156.77 | loss_wt=278.13 | loss_pct=2.49 kname=SAMIR DHARA AND CO. | net_wt=15392.3 | loss_wt=372.91 | loss_pct=2.42 kname=BIBHAS MAJI | net_wt=8704.66 | loss_wt=202.05 | loss_pct=2.32 ... (20 rows total)
CHECKS	loss % definition

C08 · Critical	What is the month-wise trend of finished product receive quantity and net weight?
VALIDATION SQL (MySQL 8)	SELECT DATE_FORMAT(e.voucher_date,'%Y-%m') AS month, SUM(i.qty) AS pieces_received, ROUND(SUM(i.net_wt),2) AS net_wt FROM finishproductreceivedentryitems i JOIN finishproductreceivedentries e ON e.id=i.fprentries_id GROUP BY DATE_FORMAT(e.voucher_date,'%Y-%m') ORDER BY month
EXPECTED RESULT	month=2026-08 | pieces_received=410 | net_wt=7259.81 month=2026-09 | pieces_received=6204 | net_wt=89142.65 month=2026-10 | pieces_received=3048 | net_wt=56197.81
CHECKS	month trend via entries<-items

C09 · Critical	What share of rework or rejection records does each karigar account for?
VALIDATION SQL (MySQL 8)	SELECT k.kname, COUNT(*) AS records, ROUND(100.0*COUNT(*)/(SELECT COUNT(*) FROM customers),1) AS share_pct FROM customers r JOIN karigars k ON k.kid=r.kid GROUP BY k.kid, k.kname ORDER BY records DESC LIMIT 10
EXPECTED RESULT	kname=SAMIR DHARA AND CO. | records=68 | share_pct=21.9 kname=BHAGYA LAXMI JEWELLERS | records=49 | share_pct=15.8 kname=ABHISHEK DAS | records=36 | share_pct=11.6 kname=BALARAM PORE | records=33 | share_pct=10.6 kname=HMR JEWELS | records=19 | share_pct=6.1 kname=KAMAL PAUL | records=18 | share_pct=5.8 kname=SAMIR SAMANTA | records=17 | share_pct=5.5 kname=BIBHAS MAJI | records=12 | share_pct=3.9 kname=BISWANATH BERA | records=10 | share_pct=3.2 kname=SHRI GANNAYAK JEWELLERS | records=9 | share_pct=2.9
CHECKS	share % rework

C10 · Critical	What share of quality check items is still in Processing for each karigar?
VALIDATION SQL (MySQL 8)	SELECT k.kname, COUNT(*) AS qc_items, SUM(q.status='Processing') AS processing_items, ROUND(100.0*SUM(q.status='Processing')/COUNT(*),1) AS processing_pct FROM qualitycheckitems q JOIN karigars k ON k.id=q.karigar_id GROUP BY k.id, k.kname ORDER BY processing_pct DESC, processing_items DESC
EXPECTED RESULT	kname=AMAL ROY | qc_items=54 | processing_items=12 | processing_pct=22.2 kname=SHRI GANNAYAK JEWELLERS | qc_items=488 | processing_items=79 | processing_pct=16.2 kname=HMR JEWELS | qc_items=326 | processing_items=40 | processing_pct=12.3 kname=KAMAL PAUL | qc_items=425 | processing_items=43 | processing_pct=10.1 kname=SUBHAMAY KARMAKAR | qc_items=287 | processing_items=27 | processing_pct=9.4 kname=JITEN PAUL | qc_items=386 | processing_items=27 | processing_pct=7.0 kname=ABHISHEK DAS | qc_items=1948 | processing_items=131 | processing_pct=6.7 kname=KAJAL KHAMRAI | qc_items=48 | processing_items=3 | processing_pct=6.3 kname=BHAGYA LAXMI JEWELLERS | qc_items=2582 | processing_items=153 | processing_pct=5.9 kname=BIBHAS MAJI | qc_items=484 | processing_items=20 | processing_pct=4.1 ... (20 rows total)
CHECKS	QC processing share (karigar_id=karigars.id)

C11 · Critical	How many issued jobs have had no quality check yet, and how many jobs passed quality check but have no finished product receive?
VALIDATION SQL (MySQL 8)	SELECT (SELECT COUNT(DISTINCT job_no) FROM issuetokarigaritems WHERE job_no NOT IN (SELECT job_no FROM qualitycheckitems)) AS issued_no_qc, (SELECT COUNT(DISTINCT job_no) FROM qualitycheckitems WHERE job_no NOT IN (SELECT job_no FROM finishproductreceivedentryitems)) AS qc_no_receive
EXPECTED RESULT	issued_no_qc=116 | qc_no_receive=4
CHECKS	stage gaps by job_no

C12 · Critical	Which job orders have not been issued to any karigar yet, and what open quantity do they carry?
VALIDATION SQL (MySQL 8)	SELECT COUNT(*) AS orders_not_issued, SUM(x.qty) AS open_qty FROM (SELECT o.id, SUM(c.ord_qty_actual) AS qty FROM customerorders o JOIN customerorderitems c ON c.order_id=o.id WHERE NOT EXISTS (SELECT 1 FROM issuetokarigars g WHERE g.order_id=o.id) GROUP BY o.id) x
EXPECTED RESULT	orders_not_issued=13 | open_qty=220
CHECKS	orders not yet issued

C13 · Critical	What is the split of orders by type (Customer vs Regular) and how many of each are still Processing?
VALIDATION SQL (MySQL 8)	SELECT type, COUNT(*) AS orders, SUM(issue_to_karigar='Processing') AS processing, ROUND(100.0*SUM(issue_to_karigar='Processing')/COUNT(*),1) AS processing_pct FROM customerorders GROUP BY type ORDER BY orders DESC
EXPECTED RESULT	type=Customer | orders=439 | processing=24 | processing_pct=5.5 type=Regular | orders=140 | processing=13 | processing_pct=9.3
CHECKS	orders by type

C14 · Critical	What are the top 10 item descriptions by total ordered quantity?
VALIDATION SQL (MySQL 8)	SELECT TRIM(description) AS description, SUM(ord_qty_actual) AS ordered_qty, COUNT(*) AS lines FROM customerorderitems GROUP BY TRIM(description) ORDER BY ordered_qty DESC, lines DESC LIMIT 10
EXPECTED RESULT	description=FINGER RING | ordered_qty=966 | lines=184 description=PENDENT | ordered_qty=910 | lines=67 description=22KT - BANGLE | ordered_qty=866 | lines=453 description=22KT - PENDENT | ordered_qty=487 | lines=139 description=22KT - JHUMAKA | ordered_qty=340 | lines=26 description=22KT - HALF SET | ordered_qty=318 | lines=97 description=BANGLE | ordered_qty=270 | lines=147 description=DROP EARRING | ordered_qty=266 | lines=26 description=JHUMAKA | ordered_qty=240 | lines=21 description=PENDANT & E_RING | ordered_qty=229 | lines=36
CHECKS	TRIM description grouping

C15 · Critical	Which 5 item descriptions have the highest total ordered weight, and what is their share of total ordered weight?
VALIDATION SQL (MySQL 8)	SELECT TRIM(description) AS description, ROUND(SUM(total_wt),2) AS total_weight, ROUND(100.0*SUM(total_wt)/(SELECT SUM(total_wt) FROM customerorderitems),1) AS share_pct FROM customerorderitems GROUP BY TRIM(description) ORDER BY total_weight DESC LIMIT 5
EXPECTED RESULT	description=22KT - BANGLE | total_weight=26286.93 | share_pct=18.3 description=22KT - HALF SET | total_weight=10682.53 | share_pct=7.4 description=BANGLE | total_weight=6265.79 | share_pct=4.4 description=PENDENT | total_weight=4375.56 | share_pct=3.0 description=22KT - PENDENT WITH CHAIN | total_weight=4180.68 | share_pct=2.9
CHECKS	weight share

C16 · Critical	Compare metal issued to each karigar with finished product net weight received from them.
VALIDATION SQL (MySQL 8)	SELECT k.kname, ROUND(m.issued,2) AS metal_issued, ROUND(f.received,2) AS fp_net_wt_received, ROUND(m.issued-f.received,2) AS difference FROM karigars k JOIN (SELECT karigar_id, SUM(weight) AS issued FROM metalissueentries GROUP BY karigar_id) m ON m.karigar_id=k.id LEFT JOIN (SELECT karigar_id, SUM(voucher_net_wt) AS received FROM finishproductreceivedentries GROUP BY karigar_id) f ON f.karigar_id=k.id ORDER BY difference DESC
EXPECTED RESULT	kname=AMAL ROY | metal_issued=1552.74 | fp_net_wt_received=993.88 | difference=558.86 kname=GITA DHAR | metal_issued=982.26 | fp_net_wt_received=1250.69 | difference=-268.43 kname=KAJAL KHAMRAI | metal_issued=1152.39 | fp_net_wt_received=1514.65 | difference=-362.26 kname=PALASH DAS JEWELLERS | metal_issued=2004.98 | fp_net_wt_received=2376.45 | difference=-371.47 kname=CHAND RATAN SONI | metal_issued=54.97 | fp_net_wt_received=484.47 | difference=-429.5 kname=BISWANATH BERA | metal_issued=3457.47 | fp_net_wt_received=4002.47 | difference=-545.0 kname=SAROJ KUMAR SAMANTA | metal_issued=3178.89 | fp_net_wt_received=4009.28 | difference=-830.39 kname=SANTU KUMAR MANNA | metal_issued=3608.48 | fp_net_wt_received=5380.14 | difference=-1771.66 kname=RANJIT SAMANTA | metal_issued=2443.83 | fp_net_wt_received=4314.83 | difference=-1871.0 kname=BIBHAS MAJI | metal_issued=6664.61 | fp_net_wt_received=8704.66 | difference=-2040.05 ... (20 rows total)
CHECKS	metal issue vs FP receive

C17 · Critical	Which karigars have a negative metal balance and how much?
VALIDATION SQL (MySQL 8)	SELECT kname, kid, ROUND(col_bal,3) AS balance FROM karigars WHERE col_bal<0 ORDER BY col_bal
EXPECTED RESULT	kname=BIBHAS MAJI | kid=BM | balance=-625.583 kname=SHRI GANNAYAK JEWELLERS | kid=SGJ | balance=-400.569 kname=SUBHAMAY KARMAKAR | kid=SK | balance=-14.362
CHECKS	karigars.col_bal

C18 · Critical	Which karigars have the highest rejected finished product percentage?
VALIDATION SQL (MySQL 8)	SELECT k.kname, COUNT(*) AS items_received, SUM(i.deliverery_status='reject') AS rejected_items, ROUND(100.0*SUM(i.deliverery_status='reject')/COUNT(*),2) AS reject_pct FROM finishproductreceivedentryitems i JOIN finishproductreceivedentries e ON e.id=i.fprentries_id JOIN karigars k ON k.id=e.karigar_id GROUP BY k.id, k.kname HAVING COUNT(*)>=50 ORDER BY reject_pct DESC LIMIT 5
EXPECTED RESULT	kname=KAMAL PAUL | items_received=438 | rejected_items=17 | reject_pct=3.88 kname=SAMIR SAMANTA | items_received=320 | rejected_items=12 | reject_pct=3.75 kname=SANTU KUMAR MANNA | items_received=542 | rejected_items=17 | reject_pct=3.14 kname=SAMIR DHARA AND CO. | items_received=649 | rejected_items=12 | reject_pct=1.85 kname=SUBHAMAY KARMAKAR | items_received=268 | rejected_items=4 | reject_pct=1.49
CHECKS	reject % (deliverery_status)

C19 · Critical	Which karigars have the most rework records per 100 finished pieces received?
VALIDATION SQL (MySQL 8)	SELECT k.kname, r.rework_records, f.pieces, ROUND(100.0*r.rework_records/f.pieces,2) AS rework_per_100 FROM karigars k JOIN (SELECT kid, COUNT(*) AS rework_records FROM customers GROUP BY kid) r ON r.kid=k.kid JOIN (SELECT kid, SUM(qty) AS pieces FROM finishproductreceivedentryitems GROUP BY kid) f ON f.kid=k.kid ORDER BY rework_per_100 DESC LIMIT 5
EXPECTED RESULT	kname=BALARAM PORE | rework_records=33 | pieces=282 | rework_per_100=11.7 kname=SAMIR DHARA AND CO. | rework_records=68 | pieces=649 | rework_per_100=10.48 kname=KAJAL KHAMRAI | rework_records=3 | pieces=48 | rework_per_100=6.25 kname=HMR JEWELS | rework_records=19 | pieces=305 | rework_per_100=6.23 kname=SAMIR SAMANTA | rework_records=17 | pieces=320 | rework_per_100=5.31
CHECKS	rework per 100 pieces

C20 · Critical	How many finished product items have loss percentage above 4, and which karigars account for them?
VALIDATION SQL (MySQL 8)	SELECT k.kname, COUNT(*) AS items_over_4pct, ROUND(SUM(i.loss_wt),2) AS loss_wt FROM finishproductreceivedentryitems i JOIN karigars k ON k.kid=i.kid WHERE i.loss_percentage>4 GROUP BY k.kid, k.kname ORDER BY items_over_4pct DESC LIMIT 10
EXPECTED RESULT	kname=KAMAL PAUL | items_over_4pct=26 | loss_wt=33.19 kname=SANTU KUMAR MANNA | items_over_4pct=1 | loss_wt=0.43
CHECKS	loss_percentage filter

C21 · Critical	What is the fulfilment percentage (finished pieces received vs quantity issued) per karigar?
VALIDATION SQL (MySQL 8)	SELECT k.kname, a.issued_qty, COALESCE(b.received_qty,0) AS received_qty, ROUND(100.0*COALESCE(b.received_qty,0)/a.issued_qty,1) AS fulfilment_pct FROM karigars k JOIN (SELECT kid, SUM(qty) AS issued_qty FROM issuetokarigaritems GROUP BY kid) a ON a.kid=k.kid LEFT JOIN (SELECT kid, SUM(qty) AS received_qty FROM finishproductreceivedentryitems GROUP BY kid) b ON b.kid=k.kid ORDER BY fulfilment_pct ASC LIMIT 10
EXPECTED RESULT	kname=MISMATCH | issued_qty=90 | received_qty=0 | fulfilment_pct=0.0 kname=AMAL ROY | issued_qty=67 | received_qty=44 | fulfilment_pct=65.7 kname=CHAND RATAN SONI | issued_qty=7 | received_qty=6 | fulfilment_pct=85.7 kname=SUBHAMAY KARMAKAR | issued_qty=295 | received_qty=268 | fulfilment_pct=90.8 kname=PALASH DAS JEWELLERS | issued_qty=105 | received_qty=104 | fulfilment_pct=99.0 kname=SHRI GANNAYAK JEWELLERS | issued_qty=421 | received_qty=417 | fulfilment_pct=99.0 kname=RANJIT SAMANTA | issued_qty=583 | received_qty=592 | fulfilment_pct=101.5 kname=ABHISHEK DAS | issued_qty=1822 | received_qty=1866 | fulfilment_pct=102.4 kname=BHAGYA LAXMI JEWELLERS | issued_qty=2409 | received_qty=2486 | fulfilment_pct=103.2 kname=JITEN PAUL | issued_qty=347 | received_qty=364 | fulfilment_pct=104.9
CHECKS	fulfilment %

C22 · Critical	How many jobs were finally received after their issue due date, and what is the average delay in days?
VALIDATION SQL (MySQL 8)	SELECT COUNT(*) AS late_jobs, ROUND(AVG(DATEDIFF(r.last_received, d.due)),1) AS avg_delay_days FROM (SELECT job_no, MIN(delivery_date) AS due FROM issuetokarigaritems GROUP BY job_no) d JOIN (SELECT f.job_no, MAX(e.voucher_date) AS last_received FROM finishproductreceivedentryitems f JOIN finishproductreceivedentries e ON e.id=f.fprentries_id GROUP BY f.job_no) r ON r.job_no=d.job_no WHERE r.last_received > d.due
EXPECTED RESULT	late_jobs=321 | avg_delay_days=24.7
CHECKS	due vs received delay (date)

C23 · Critical	What is the company metal stock position: total received, total consumed (issue_qty) and balance?
VALIDATION SQL (MySQL 8)	SELECT ROUND(SUM(weight),3) AS total_received, ROUND(SUM(issue_qty),3) AS total_issued_out, ROUND(SUM(balance_qty),3) AS balance FROM metalreceiveentries
EXPECTED RESULT	total_received=146892.733 | total_issued_out=87247.649 | balance=59645.084
CHECKS	metal stock

C24 · Critical	Which purity has the most metal issued to karigars (by weight)?
VALIDATION SQL (MySQL 8)	SELECT p.purity, ROUND(SUM(m.weight),2) AS total_weight, COUNT(*) AS entries FROM metalissueentries m JOIN metalpurities p ON p.purity_id=m.purity_id GROUP BY p.purity_id, p.purity ORDER BY total_weight DESC LIMIT 5
EXPECTED RESULT	purity=99.5 | total_weight=46599.37 | entries=52 purity=99.99 | total_weight=39814.63 | entries=40 purity=91.6 | total_weight=706.55 | entries=11 purity=75.0 | total_weight=127.1 | entries=3
CHECKS	metal purity join

C25 · Critical	Trace job 11739268 through the pipeline: ordered, issued, quality checked, finished received and finished goods voucher quantities.
VALIDATION SQL (MySQL 8)	SELECT (SELECT SUM(c.ord_qty_actual) FROM customerorderitems c JOIN customerorders o ON o.id=c.order_id WHERE o.jo_no='11739268') AS ordered_qty, (SELECT SUM(qty) FROM issuetokarigaritems WHERE job_no='11739268') AS issued_qty, (SELECT COUNT(*) FROM qualitycheckitems WHERE job_no='11739268') AS qc_items, (SELECT COUNT(*) FROM finishproductreceivedentryitems WHERE job_no='11739268') AS fp_received_items, (SELECT COUNT(*) FROM f_g_in_out__d_e_t_a_i_l WHERE job_no='11739268') AS fg_lines
EXPECTED RESULT	ordered_qty=807 | issued_qty=808 | qc_items=912 | fp_received_items=876 | fg_lines=832
CHECKS	single job trace across 5 tables

C26 · Critical	What is the month-wise open (Processing) order quantity by delivery month?
VALIDATION SQL (MySQL 8)	SELECT DATE_FORMAT(delivery_date,'%Y-%m') AS delivery_month, COUNT(*) AS open_items, SUM(ord_qty_actual) AS open_qty FROM customerorderitems WHERE issue_to_karigar_status='Processing' AND delivery_date IS NOT NULL GROUP BY DATE_FORMAT(delivery_date,'%Y-%m') ORDER BY delivery_month
EXPECTED RESULT	delivery_month=2026-08 | open_items=16 | open_qty=159 delivery_month=2026-09 | open_items=5 | open_qty=71 delivery_month=2026-10 | open_items=1 | open_qty=2 delivery_month=2026-11 | open_items=129 | open_qty=217
CHECKS	open qty by delivery month

C27 · Critical	Which karigars have both more than 20 pending issued quantity and more than 10 rework records?
VALIDATION SQL (MySQL 8)	SELECT k.kname, p.pending_qty, r.rework_records FROM karigars k JOIN (SELECT kid, SUM(bal_qty) AS pending_qty FROM issuetokarigaritems GROUP BY kid HAVING SUM(bal_qty)>20) p ON p.kid=k.kid JOIN (SELECT kid, COUNT(*) AS rework_records FROM customers GROUP BY kid HAVING COUNT(*)>10) r ON r.kid=k.kid ORDER BY p.pending_qty DESC
EXPECTED RESULT	kname=ABHISHEK DAS | pending_qty=98 | rework_records=36 kname=BHAGYA LAXMI JEWELLERS | pending_qty=95 | rework_records=49 kname=SAMIR DHARA AND CO. | pending_qty=80 | rework_records=68 kname=KAMAL PAUL | pending_qty=36 | rework_records=18
CHECKS	two-condition HAVING cross-table

C28 · Critical	Which 5 karigars have the highest loss weight on finished goods in/out vouchers, and what is their loss percentage?
VALIDATION SQL (MySQL 8)	SELECT h.party_name AS karigar, ROUND(SUM(d.loss_wt),2) AS loss_wt, ROUND(SUM(d.net_wt),2) AS net_wt, ROUND(100.0*SUM(d.loss_wt)/SUM(d.net_wt),2) AS loss_pct FROM f_g_in_out__d_e_t_a_i_l d JOIN f_g_in_out__h_e_a_d_e_r h ON h.id=d.header_id GROUP BY h.party_id, h.party_name ORDER BY loss_wt DESC LIMIT 5
EXPECTED RESULT	karigar=BHAGYA LAXMI JEWELLERS | loss_wt=491.9 | net_wt=18051.56 | loss_pct=2.72 karigar=KAMAL PAUL | loss_wt=452.22 | net_wt=11861.98 | loss_pct=3.81 karigar=ABHISHEK DAS | loss_wt=352.73 | net_wt=12963.21 | loss_pct=2.72 karigar=SAMIR DHARA AND CO. | loss_wt=270.55 | net_wt=13794.0 | loss_pct=1.96 karigar=HMR JEWELS | loss_wt=243.87 | net_wt=10580.37 | loss_pct=2.3
CHECKS	f_g header+detail loss

C29 · Critical	What is the share of 18KT versus 22KT in total ordered quantity and ordered weight?
VALIDATION SQL (MySQL 8)	SELECT kt, SUM(ord_qty_actual) AS ordered_qty, ROUND(100.0*SUM(ord_qty_actual)/(SELECT SUM(ord_qty_actual) FROM customerorderitems),1) AS qty_share_pct, ROUND(SUM(total_wt),2) AS ordered_wt, ROUND(100.0*SUM(total_wt)/(SELECT SUM(total_wt) FROM customerorderitems),1) AS wt_share_pct FROM customerorderitems GROUP BY kt ORDER BY kt
EXPECTED RESULT	kt=18 | ordered_qty=3276 | qty_share_pct=34.9 | ordered_wt=21921.52 | wt_share_pct=15.2 kt=22 | ordered_qty=6112 | qty_share_pct=65.1 | ordered_wt=121903.62 | wt_share_pct=84.8
CHECKS	kt share

C30 · Critical	Which karigars have never been issued any job?
VALIDATION SQL (MySQL 8)	SELECT k.kid, k.kname FROM karigars k WHERE NOT EXISTS (SELECT 1 FROM issuetokarigaritems i WHERE i.kid=k.kid) ORDER BY k.kname
EXPECTED RESULT	kid=AD | kname=ANIL KUMAR DHAR kid=AT | kname=ASISH TARAFDAR kid=XX | kname=AUTO kid=DD | kname=DINDAYAL SONI kid=SB | kname=KALA KALPANA JEWELLERS kid=LG | kname=LOKENATH GOLD kid=YS | kname=NAVLAKSHMI JEWELLERS kid=NP | kname=NIRMAL PAL kid=BG | kname=RUMPA GUCHHAIT kid=SKP | kname=SANJIB KUMAR PAUL JEWELLERS ... (14 rows total)
CHECKS	NOT EXISTS karigars